

Automated minus reference counts on the gold standard: protocol change

tranche mixed_gold_cont120_01 - frozen baseline versus candidate protocol on the same manifest and gold - misses and abstentions scored as 0 - square-root axes - every asserted gold row

● automated count supplied ○ variant matched, no automated count (scored 0) ○ variant not found (scored 0)

Marker area - gold rows at one coordinate (wedges split it by status):

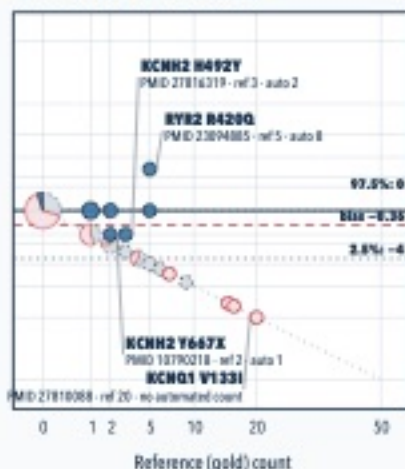
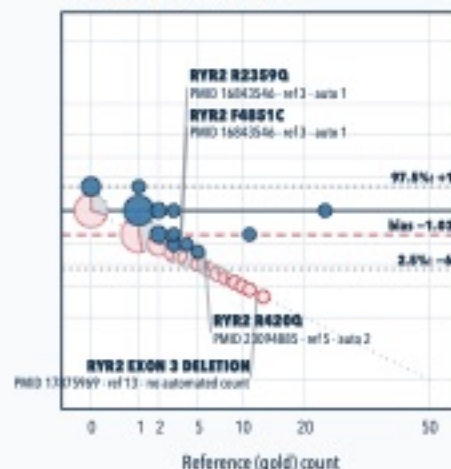
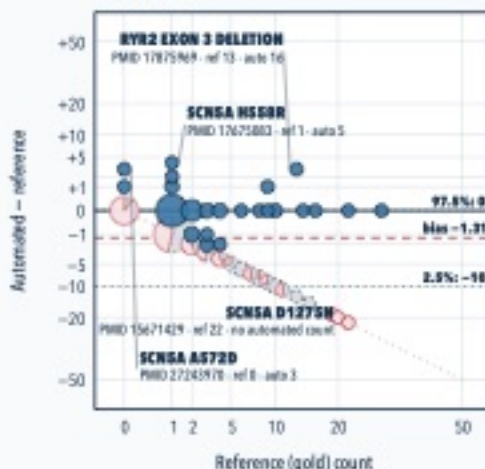
○ 1 ○ 8 ○ 28

Baseline arm - 20260903_protocol_cont120_01_baseline - runtime 81080791 - locked 2026-09-03 - 120 gene-paper attempts

Carriers

Affected individuals

Unaffected individuals



384 gold rows - matched 68.5% - supplied 32.0%
exact 40.6% (zero-baseline 13.0%) - within ± 1 81.8%
bias -1.31 - c2e MAE 1.393 - cond. MAE 0.236 (n 123)
empirical 95% limits -10 to 0 - RMSE 3.05

383 gold rows - matched 68.4% - supplied 23.0%
exact 37.9% (zero-baseline 21.1%) - within ± 1 83.8%
bias -1.03 - c2e MAE 1.091 - cond. MAE 0.330 (n 88)
empirical 95% limits -6 to +1 - RMSE 1.98

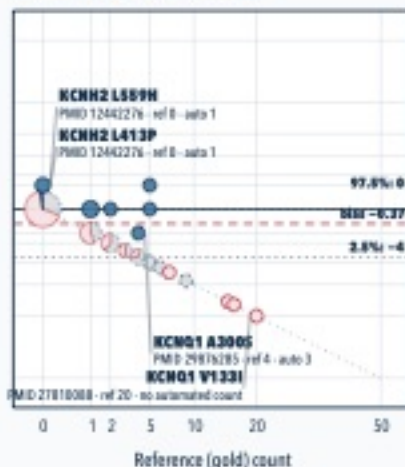
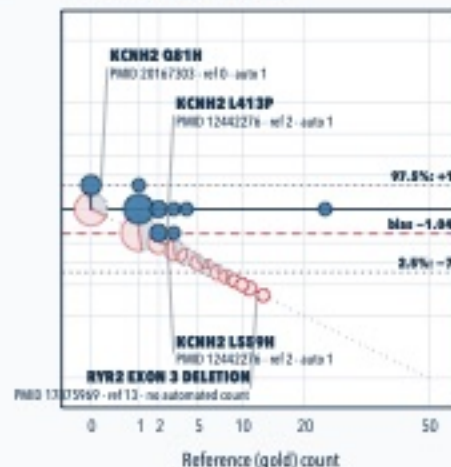
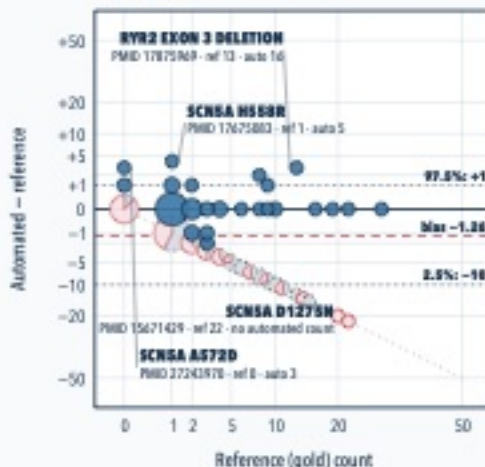
383 gold rows - matched 66.4% - supplied 7.3%
exact 88.8% (zero-baseline 82.2%) - within ± 1 94.8%
bias -0.36 - c2e MAE 0.379 - cond. MAE 0.179 (n 28)
empirical 95% limits -4 to 0 - RMSE 1.80

Candidate arm - 20260903_protocol_cont120_01_candidate - runtime d99724bf - locked 2026-09-03 - 120 gene-paper attempts

Carriers

Affected individuals

Unaffected individuals



384 gold rows - matched 68.8% - supplied 33.9%
exact 41.4% (zero-baseline 12.8%) - within ± 1 81.8%
bias -1.26 - c2e MAE 1.359 - cond. MAE 0.231 (n 136)
empirical 95% limits -10 to +1 - RMSE 2.96

383 gold rows - matched 67.9% - supplied 23.8%
exact 37.9% (zero-baseline 20.1%) - within ± 1 84.6%
bias -1.04 - c2e MAE 1.117 - cond. MAE 0.253 (n 91)
empirical 95% limits -7 to +1 - RMSE 2.07

383 gold rows - matched 67.9% - supplied 5.7%
exact 87.3% (zero-baseline 83.3%) - within ± 1 93.7%
bias -0.37 - c2e MAE 0.386 - cond. MAE 0.227 (n 22)
empirical 95% limits -4 to 0 - RMSE 1.80

Candidate - baseline on the same 1150 gold rows (paired by gold row; deltas are candidate minus baseline)

Carriers

Affected individuals

Unaffected individuals

c2e MAE 1.393 $\rightarrow$ 1.359 (-0.034) - registered UB +0.083
supplied 123 $\rightarrow$ 130 - exact 156 $\rightarrow$ 159 - bias -1.31 $\rightarrow$ -1.26
rows better 13 - worse 10 - unchanged 361
coverage on matched 46.8% $\rightarrow$ 49.8% (+3.04 pp) - LB -1.67 pp

c2e MAE 1.091 $\rightarrow$ 1.117 (+0.026)
supplied 88 $\rightarrow$ 91 - exact 145 $\rightarrow$ 145 - bias -1.03 $\rightarrow$ -1.04
rows better 15 - worse 19 - unchanged 349
coverage on matched 33.6% $\rightarrow$ 35.0% (+1.41 pp)

c2e MAE 0.379 $\rightarrow$ 0.386 (+0.008)
supplied 28 $\rightarrow$ 22 - exact 340 $\rightarrow$ 336 - bias -0.36 $\rightarrow$ -0.37
rows better 3 - worse 7 - unchanged 373
coverage on matched 18.7% $\rightarrow$ 8.5% (-2.23 pp)

Identity (primary endpoint): recall 68.49% $\rightarrow$ 67.97% (-0.52 pp; LB -1.40 pp) - precision 65.59% $\rightarrow$ 64.93% (-0.66 pp; LB -2.60 pp) - registered decision: reject_or_revise_candidate - secondary counts